

Review game files

09/30/2026 22:13:04

<https://chatgpt.com/c/6abc7f72-9ca8-83e9-b399-be0fa9e83628>

main(5).py README (1).md game_engine.py board.py These are my current files

Got them. I have the current versions of: main.py game_engine.py board.py README.md I can see the proje...

I am working on this Python Pygame Match-3 game. Please inspect the existing code carefully before makin...

I inspected the existing board.py rather than changing any game logic. The crash is caused by...

I need you to fix the existing code carefully. I previously asked you to fix the NameError involving variable `c...

board(3).py I need you to fix the existing code carefully. I previously asked you to fix the NameError involvin...

I checked the attached board.py line-by-line. Importantly, the missing c loop is already present in this versio...

This is the error C:\Users\sufiy\57_match_three_gem>python main.py pygame 2.6.1 (SDL 2.28.4, Python...

Yes - this traceback confirms the problem exactly. The error is still: NameError: name 'c' is not defined at: if...

I am working on a Python Pygame Match-3 game. When I run `python main.py`, the game sometimes crash...

The cause is very specific: in drop_and_refill(), the variable c is used, but there is no loop defining c. Your...

I am working on a Python Pygame Match-3 game. I have already fixed a startup bug in `game/board.py`...

The bug is in process_swap(). The move is being decremented before the code checks whether the swap...

okay now for the same code I am working on a Python Pygame Match-3 game. I have already fixed two bugs...

Thinking ● ✓ Worked for 9s Yes. The existing structure already makes this fairly simple because...

I am working on a Python Pygame Match-3 game. I have already completed these tasks: 1. Fixed the `c`...

Yes. Since you already have the c fix, invalid-swap fix, and cascade scoring, we can add the Bomb Gem with...

C:\Users\sufiy\57_match_three_gem>python main.py pygame 2.6.1 (SDL 2.28.4, Python 3.13.1) Hello from t...

Yep — this error is caused by the change I suggested to find_matches(). The problem is that we changed:...

Pasted code.py I am working on a Python Pygame Match-3 game. I have already completed these tasks: 1....

I inspected the current board.py. The existing game loop already calls Board.update() and Board.render(), so...

is the indexing correct? def render(self, surface): board_rect = pygame.Rect(self.offset_x, self.offset_y,...

Almost — the (r, c) indexing is correct, but there is one important issue with tile_rect. Your indexing This is...

now ? def render(self, surface): board_rect = pygame.Rect(self.offset_x, self.offset_y, GRID_SIZE * TILE_SIZE...

Yes — now the indexing and placement are correct. ✓ Your structure is right: gem = self.grid[r][c] and: if...



main(5).py



README (1).md



game_engine.py



board.py

These are my current files

9/30/2026, 8:48:20 AM



Got them. I have the current versions of:

- `main.py`
- `game_engine.py`
- `board.py`
- `README.md`

I can see the project is the **Match-3 Gem Swap Repair Lab**, with the four required tasks: invalid-move deduction, cascade scoring, 4-in-a-row bomb gems, and idle hints. `README (1).md`

I also see the current intentional bug in `process_swap()` : the move is deducted **before** checking whether the swap produced a match. `board.py`

And your `main.py` currently imports `GameEngine` from `game.game_engine` , so the expected folder structure is `game/main.py` with `board.py` and `game_engine.py` inside `game/` . `main.py`

Send me what you want to do next - for example, **fix Task 1**, implement **all 4 tasks**, or troubleshoot the error you were getting when running `main.py` .

9/30/2026, 8:48:23 AM · gpt-5-6





I am working on this Python Pygame Match-3 game. Please inspect the existing code carefully before making changes.

When I run:

```
python main.py
```

the game crashes during board initialization with this error:

NameError: name 'c' is not defined

The traceback points to:

game/board.py → drop_and_refill() → line 127

First identify why variable `c` is undefined in `drop\_and\_refill()` and explain the bug.

Then fix ONLY this startup/runtime error. Do not implement any of the additional features from the README yet, and do not change unrelated game behavior.

After making the change, tell me exactly what was changed and how I should test it.

9/30/2026, 8:48:42 AM



I inspected the existing `board.py` rather than changing any game logic. The crash is caused by `drop_and_refill()` using `c` without ever creating a column loop.

The minimal fix is to add the missing `for c in range(GRID_SIZE)` loop and indent the existing refill logic under it. I have not touched swap scoring, bomb gems, cascades, hints, or other behavior.

The edited file passes Python compilation. I couldn't run the Pygame initialization in this environment because `pygame` isn't installed here, so your local `python main.py` test is still the definitive runtime check.

Thinking

- 🌐 **Fixing board drop and refill logic**
 - **Fixed board drop and refill logic**
- 🌐 **Verifying Board Syntax and Initialization**
 - **Verified board syntax and initialization**
- 🌐 **Inspecting Corrected Board File**
 - **Inspecting Corrected Board File**

✓ Worked for 21s

Why the error happens

In your original `drop_and_refill()`, the code uses `c` here:

```
if self.grid[r][c] is None:
```

but there was **no** `for c ...` **loop** defining `c`. The function only had a loop over `r`. `board.py`

This means Python reaches `self.grid[r][c]` and asks, essentially, “What is `c`?” - but `c` doesn't exist in that scope, producing:

```
NameError: name 'c' is not defined
```

Exactly what I changed

I added the missing column loop:

```
def drop_and_refill(self):
    for c in range(GRID_SIZE):
        empty_slots = 0
        for r in range(GRID_SIZE - 1, -1, -1):
            ...
```

So now the function processes **each column** `c` **from 0 to 7**, and within each column processes the rows from bottom to top.

Nothing else was changed. In particular:

- ❌ No cascade scoring
- ❌ No bomb gems
- ❌ No idle hints
- ❌ No changes to move deduction
- ❌ No changes to matching behavior
- ❌ No changes to scoring
- ✅ Only the undefined-variable/runtime error was fixed

The corrected section is now structured as expected. `board.py`

Test it

Replace your local `game/board.py` with the corrected file:

[Download the corrected `board.py`](#)

Then from your project root run:

```
python main.py
```

You should now get the Pygame window instead of the `NameError`.

Then verify:

1. The **8×8 board appears**.
2. Gems are displayed normally.
3. You can click two adjacent gems.
4. Matching gems still clear and refill.
5. Press `R` and confirm the board resets.
6. Don't worry about the README's additional tasks yet-we have intentionally left them untouched.

I also verified that the modified `board.py` passes Python's syntax/compilation check. The only test I couldn't perform here was actually launching Pygame because this execution environment doesn't have `pygame` installed.

9/30/2026, 8:48:47 AM · gpt-5-6



I need you to fix the existing code carefully.

I previously asked you to fix the `NameError` involving variable ``c`` in ``game/board.py -> drop_and_refill()``, but the resulting code is unreliable: the game only opens intermittently and often crashes again when I run it.

I have attached the current `board.py`.

IMPORTANT:

- Do NOT rewrite the entire file.
- Do NOT redesign the game.
- Do NOT add any of the README features yet.
- Preserve the existing game architecture and behavior.
- Make the smallest possible change needed to correctly fix the startup/runtime error.

First inspect the current ``drop_and_refill()`` implementation and determine exactly why the original ``NameError: name 'c' is not defined`` occurred.

Then inspect the surrounding initialization and ``resolve_matches()`` flow to make sure the fix does not introduce another runtime problem.

After identifying the problem, make ONLY the necessary code change.

Then explain:

1. What caused the original `NameError`.
2. What exact lines you changed.
3. Why the new code is safe.
4. How I can test it repeatedly by running ``python main.py`` multiple times.

Do not modify ``process_swap()``, scoring, rendering, or any other functionality in this step.

9/30/2026, 7:18:24 PM



board(3).py

I need you to fix the existing code carefully.

I previously asked you to fix the `NameError` involving variable ``c`` in ``game/board.py -> drop_and_refill()``, but the resulting code is unreliable: the game only opens intermittently and often crashes again when I run it.

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Do not modify ``process_swap()``, scoring, rendering, or any other functionality in this step.

9/30/2026, 7:18:48 PM



I checked the attached `board.py` line-by-line. Importantly, the missing `c` loop is **already present** in this version, so I'm not going to make a duplicate or unrelated edit. I'm now checking whether the surrounding initialization can repeatedly enter `resolve_matches()` safely.

Thinking

- **Reviewing the loop fix**

I'm checking the current code: the loop already defines ``c``, so the reported ``NameError`` appears resolved. I'm also watching for a separate startup hang caused by repeated random refills during match resolution, but I'm not changing unrelated behavior.

- **Reviewed the loop fix**



- **Verifying match resolution and startup stability**

- **Validating Python file syntax**



✓ Worked for 23s

I inspected the **attached current** `board.py` carefully. The important finding is that the `NameError` fix is **already correctly present** in the file you just uploaded, so making another change would actually violate your requirement to make only the necessary change.

1. What caused the original `NameError` ?

Originally, `drop_and_refill()` used:

```
self.grid[r][c]
```

but there was no loop defining `c`.

The function was effectively doing:

```
for r in range(GRID_SIZE - 1, -1, -1):
    if self.grid[r][c] is None:
```

`r` existed because of the loop, but `c` did not. Therefore Python raised:

```
NameError: name 'c' is not defined
```

The correct fix is to process the board **column by column**.

Your current file now has exactly that:

```
def drop_and_refill(self):
    for c in range(GRID_SIZE):
        empty_slots = 0
        for r in range(GRID_SIZE - 1, -1, -1):
```

`board.py`

2. What exact lines were changed?

In the current attached file, the necessary change is:

```
def drop_and_refill(self):
    for c in range(GRID_SIZE):
        empty_slots = 0
        for r in range(GRID_SIZE - 1, -1, -1):
```

The crucial addition is:

```
for c in range(GRID_SIZE):
```

and the existing refill code is inside that loop.

There is nothing further I would change in `drop_and_refill()` right now.

3. Why is this safe?

Look at the initialization flow:

```
Board.__init__()
↓
reset()
↓
```

```
fill 8x8 grid
↓
resolve_matches()
↓
find_matches()
↓
remove matched gems
↓
drop_and_refill()
```

Your `reset()` calls `resolve_matches()` immediately after filling the grid. `board.py`

When `resolve_matches()` finds a match, it sets those positions to `None` and then calls `drop_and_refill()`. `board.py`

With:

```
for c in range(GRID_SIZE):
```

`c` is guaranteed to be `0` through `7` whenever this code runs:

```
self.grid[r][c]
```

So the original `NameError` cannot occur from this function anymore.

I also checked the attached file for Python syntax, and it parses successfully.

One important point about the "intermittent" behavior

There is **no intermittent `c`-variable bug in the current attached file**. The `c` issue is fixed deterministically.

If you're still seeing the `NameError` sometimes, the most likely explanation is that you're sometimes running a **different copy/version of `board.py`** than the one you attached-for example, an old copy in another project folder or an unsaved VS Code version.

Don't modify the code further yet. First test the exact file you're actually running.

4. Test it repeatedly

From the folder containing your `main.py`, run:

```
python main.py
```

Close the game, then run it again:

```
python main.py
```

Repeat this around **10 times**.

Also check which Python file is actually being imported. From your project folder:

```
python -c "import game.board; print(game.board.__file__)"
```

It should print the path to the `game\board.py` that you're editing.

Then run:

```
python main.py
```

If it crashes again, **send me the complete new traceback** rather than changing anything else. That will tell us whether we're dealing with a different runtime problem rather than the original `c` error.

And I have **not modified** `process_swap()`, **scoring, rendering, or any README feature**. The attached file already contains the minimal `c`-loop correction we needed.

9/30/2026, 7:18:52 PM · gpt-5-6



```

This is the error
C:\Users\sufiy\57_match_three_gem>python main.py
pygame 2.6.1 (SDL 2.28.4, Python 3.13.1)
Hello from the pygame community. https://www.pygame.org/contribute.html
Traceback (most recent call last):
  File "C:\Users\sufiy\57_match_three_gem\main.py", line 38, in <module>
    main()
    ~~~~^^
  File "C:\Users\sufiy\57_match_three_gem\main.py", line 24, in main
    engine.handle_click(event.pos)
    ~~~~~^
  File "C:\Users\sufiy\57_match_three_gem\game\game_engine.py", line 36, in
handle_click
    self.board.process_swap(prev_selected, (row, col))
    ~~~~~^
  File "C:\Users\sufiy\57_match_three_gem\game\board.py", line 169, in
process_swap
    cleared = self.resolve_matches()
    ~~~~~^
  File "C:\Users\sufiy\57_match_three_gem\game\board.py", line 151, in
resolve_matches
    self.drop_and_refill()
    ~~~~~^
  File "C:\Users\sufiy\57_match_three_gem\game\board.py", line 127, in
drop_and_refill
    if self.grid[r][c] is None:
       ^
NameError: name 'c' is not defined

```

9/30/2026, 7:21:02 PM



Yes - this traceback confirms the problem exactly.

The error is still:

```
NameError: name 'c' is not defined
```

at:

```
if self.grid[r][c] is None:
```

What is wrong

Your `drop_and_refill()` function is using `c` as the column number, but your **actual** `board.py` **that Python is running does not define** `c` before this line.

It needs to loop through all 8 columns.

Replace your **entire** `drop_and_refill()` **function** with this:

```

def drop_and_refill(self):
    for c in range(GRID_SIZE):
        empty_slots = 0

    for r in range(GRID_SIZE - 1, -1, -1):
        if self.grid[r][c] is None:

```

```

        empty_slots += 1
    elif empty_slots > 0:
        gem = self.grid[r][c]
        gem.target_row = r + empty_slots
        gem.target_y = (r + empty_slots) * TILE_SIZE

        self.grid[r + empty_slots][c] = gem
        self.grid[r][c] = None

    for r in range(empty_slots):
        color = random.choice(GEM_COLORS)
        gem = Gem(color, r, c)
        gem.current_y = -((empty_slots - r) * TILE_SIZE)
        self.grid[r][c] = gem

```

The **critical line** is:

```
for c in range(GRID_SIZE):
```

Your current function is missing that line. In the uploaded version I inspected earlier, that loop was already present, which means the file you are actually running at:

```
C:\Users\sufiy\57_match_three_gem\game\board.py
```

is different from that uploaded version. `board.py`

Why this fixes it

Think of the board as:

```

    c = 0  1  2  3  4  5  6  7
r = 0
r = 1
r = 2
...
r = 7

```

You already have:

```
for r in range(GRID_SIZE - 1, -1, -1):
```

which goes through the **rows**.

You also need:

```
for c in range(GRID_SIZE):
```

which goes through the **columns**.

So the structure is:

```

for c in range(GRID_SIZE):          # column
    for r in range(GRID_SIZE - 1, -1, -1): # row

```

Without the outer loop, Python reaches:

```
self.grid[r][c]
```

and says: "What is `c`? You never created it."

After saving `board.py`

Run:

```
python main.py
```

Then actually make a swap that causes gems to disappear. The traceback shows that the crash happens **after a successful match**, not during initial startup:

```
process_swap()  
→ resolve_matches()  
→ drop_and_refill()  
→ NameError
```

So the game can open normally and only crash when `drop_and_refill()` is triggered.

Also, don't change `process_swap()` or implement any of the README's optional features yet. This fix is only for the `c` error.

9/30/2026, 7:21:06 PM · gpt-5-6



I am working on a Python Pygame Match-3 game.

When I run `python main.py`, the game sometimes crashes during board initialization with:

NameError: name 'c' is not defined

The traceback points to:

game/board.py → drop_and_refill()

Specifically, the failing line is:

if self.grid[r][c] is None:

Please inspect the existing `drop\_and\_refill()` function carefully and identify why `c` is undefined at this point.

Fix ONLY this bug.

Requirements:

- Make the smallest possible change.
- Ensure `c` is correctly defined/scoped before it is used.
- Preserve the existing game logic.
- Do NOT rewrite the entire `board.py`.
- Do NOT change scoring, swapping, matching, UI, controls, or any other functionality.
- Do NOT add new features.
- Do NOT modify other files unless absolutely necessary for this specific bug.
- After making the fix, briefly explain what caused the error and exactly what you changed.

here is the original code

```
import random
import pygame
```

```
GRID_SIZE = 8
TILE_SIZE = 60
GEM_COLORS = [
    (220, 50, 50), # Red
    (50, 200, 50), # Green
    (50, 100, 240), # Blue
    (240, 200, 40), # Yellow
    (180, 50, 220), # Purple
    (240, 130, 40), # Orange
]
```

```
class Gem:
```

```
    def __init__(self, color, target_row, col):
        self.color = color
        self.target_row = target_row
        self.col = col
        # Start higher up to animate falling down
        self.current_y = (target_row - 2) * TILE_SIZE
```

```

        self.target_y = target_row * TILE_SIZE
        self.fall_speed = 12.0

    def update(self):
        if self.current_y < self.target_y:
            self.current_y += self.fall_speed
        if self.current_y > self.target_y:
            self.current_y = self.target_y

    def is_animating(self):
        return self.current_y < self.target_y

class Board:
    """Manages animated gem grid, gravity drops, score, and game limits."""

    def __init__(self, offset_x, offset_y, target_score=500, max_moves=20):
        self.offset_x = offset_x
        self.offset_y = offset_y
        self.target_score = target_score
        self.max_moves = max_moves
        self.grid = [[None for _ in range(GRID_SIZE)] for _ in range(GRID_SIZE)]
        self.selected = None
        self.score = 0
        self.moves_remaining = max_moves
        self.reset()

    def reset(self):
        """Reset board grid, score, and move limits."""
        self.score = 0
        self.moves_remaining = self.max_moves
        self.selected = None
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE):
                color = random.choice(GEM_COLORS)
                gem = Gem(color, r, c)
                gem.current_y = gem.target_y # Snap instantly on initial start
                self.grid[r][c] = gem

        self.resolve_matches()

    def is_animating(self):
        """Returns True if any gem is currently dropping down."""
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE):
                if self.grid[r][c] and self.grid[r][c].is_animating():
                    return True
        return False

    def swap_gems(self, pos1, pos2):
        """Swap positions and target render coordinates of two gems."""
        r1, c1 = pos1
        r2, c2 = pos2

        g1, g2 = self.grid[r1][c1], self.grid[r2][c2]
        self.grid[r1][c1], self.grid[r2][c2] = g2, g1

        if self.grid[r1][c1]:
            self.grid[r1][c1].target_row = r1

```

```

        self.grid[r1][c1].target_y = r1 * TILE_SIZE
        self.grid[r1][c1].current_y = r1 * TILE_SIZE

    if self.grid[r2][c2]:
        self.grid[r2][c2].target_row = r2
        self.grid[r2][c2].target_y = r2 * TILE_SIZE
        self.grid[r2][c2].current_y = r2 * TILE_SIZE

    def is_adjacent(self, pos1, pos2):
        r1, c1 = pos1
        r2, c2 = pos2
        return abs(r1 - r2) + abs(c1 - c2) == 1

    def find_matches(self):
        """Scan grid for horizontal and vertical 3-in-a-row color matches."""
        matched = set()

        # Horizontal matches
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE - 2):
                if (
                    self.grid[r][c]
                    and self.grid[r][c + 1]
                    and self.grid[r][c + 2]
                    and self.grid[r][c].color == self.grid[r][c + 1].color == self.grid[r][c
+ 2].color
                ):
                    matched.update([(r, c), (r, c + 1), (r, c + 2)])

        # Vertical matches
        for r in range(GRID_SIZE - 2):
            for c in range(GRID_SIZE):
                if (
                    self.grid[r][c]
                    and self.grid[r + 1][c]
                    and self.grid[r + 2][c]
                    and self.grid[r][c].color == self.grid[r + 1][c].color == self.grid[r +
2][c].color
                ):
                    matched.update([(r, c), (r + 1, c), (r + 2, c)])

        return matched

    def drop_and_refill(self):
        empty_slots = 0
        for r in range(GRID_SIZE - 1, -1, -1):
            if self.grid[r][c] is None:
                empty_slots += 1
            elif empty_slots > 0:
                gem = self.grid[r][c]
                gem.target_row = r + empty_slots
                gem.target_y = (r + empty_slots) * TILE_SIZE
                self.grid[r + empty_slots][c] = gem
                self.grid[r][c] = None

        for r in range(empty_slots):
            color = random.choice(GEM_COLORS)
            gem = Gem(color, r, c)
            gem.current_y = -(empty_slots - r) * TILE_SIZE

```

```

        self.grid[r][c] = gem

    def resolve_matches(self):
        total_cleared = 0
        while True:
            matches = self.find_matches()
            if not matches:
                break
            total_cleared += len(matches)
            for r, c in matches:
                self.grid[r][c] = None
            self.drop_and_refill()
        return total_cleared

    def process_swap(self, pos1, pos2):
        if not self.is_adjacent(pos1, pos2) or self.is_game_over() or self.is_animating():
            return False

        self.swap_gems(pos1, pos2)
        matches = self.find_matches()

        # BUG SYMPTOM:
        # Move count decrements on EVERY swap attempt even invalid ones.
        self.moves_remaining -= 1

        if not matches:
            self.swap_gems(pos1, pos2) # Revert invalid swap
            return False

        cleared = self.resolve_matches()
        self.score += cleared * 10
        return True

    def is_game_over(self):
        return self.score >= self.target_score or self.moves_remaining <= 0

    def check_result(self):
        if self.score >= self.target_score:
            return "WIN"
        if self.moves_remaining <= 0:
            return "LOSS"
        return None

    def update(self):
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE):
                if self.grid[r][c]:
                    self.grid[r][c].update()

    def render(self, surface):
        board_rect = pygame.Rect(
            self.offset_x, self.offset_y, GRID_SIZE * TILE_SIZE, GRID_SIZE * TILE_SIZE
        )
        pygame.draw.rect(surface, (20, 22, 28), board_rect, border_radius=8)
        pygame.draw.rect(surface, (60, 65, 75), board_rect, width=3,
            border_radius=8)

        for r in range(GRID_SIZE):

```

```

for c in range(GRID_SIZE):
    gem = self.grid[r][c]
    if gem:
        x = self.offset_x + c * TILE_SIZE
        y = self.offset_y + gem.current_y
        tile_rect = pygame.Rect(x + 2, y + 2, TILE_SIZE - 4, TILE_SIZE - 4)

        pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)
        pygame.draw.rect(
            surface, (255, 255, 255), tile_rect, width=1, border_radius=10
        )

    if self.selected == (r, c):
        sel_x = self.offset_x + c * TILE_SIZE
        sel_y = self.offset_y + r * TILE_SIZE
        sel_rect = pygame.Rect(sel_x + 2, sel_y + 2, TILE_SIZE - 4,
TILE_SIZE - 4)
        pygame.draw.rect(
            surface, (255, 255, 255), sel_rect, width=4, border_radius=10
        )

```

9/30/2026, 7:39:41 PM



The cause is very specific: in `drop_and_refill()`, the variable `c` is used, but there is **no loop defining** `c`.

Your current function starts with:

```

def drop_and_refill(self):
    empty_slots = 0
    for r in range(GRID_SIZE - 1, -1, -1):
        if self.grid[r][c] is None:

```

Here `r` is defined, but `c` is not.

Minimal fix

Add **one outer loop** for the columns:

```

def drop_and_refill(self):
    for c in range(GRID_SIZE):
        empty_slots = 0
        for r in range(GRID_SIZE - 1, -1, -1):
            if self.grid[r][c] is None:
                empty_slots += 1
            elif empty_slots > 0:
                gem = self.grid[r][c]
                gem.target_row = r + empty_slots
                gem.target_y = (r + empty_slots) * TILE_SIZE
                self.grid[r + empty_slots][c] = gem
                self.grid[r][c] = None

        for r in range(empty_slots):
            color = random.choice(GEM_COLORS)
            gem = Gem(color, r, c)
            gem.current_y = -(empty_slots - r) * TILE_SIZE
            self.grid[r][c] = gem

```

Exactly what changed

Only this line was added:

```
for c in range(GRID_SIZE):
```

and the existing `empty_slots = 0` and everything below it is indented inside that loop.

So the function now works **column by column**:

```
c = 0 → process column 0  
c = 1 → process column 1  
c = 2 → process column 2  
...  
c = 7 → process column 7
```

This preserves all of your existing gravity/refill logic. Nothing else in `board.py` needs to be changed.

Do not change `process_swap()`, **scoring, matching, UI, or any other code for this bug.**

9/30/2026, 7:39:43 PM · gpt-5-6



I am working on a Python Pygame Match-3 game.

I have already fixed a startup bug in `game/board.py`. Please now fix the following existing gameplay bug:

****Bug:****

When the player swaps two adjacent gems and the swap does NOT create a match, the game correctly rejects/reverts the swap, but it still decreases `moves\_remaining`.

****Expected behavior:****

- A valid swap that creates a match should decrease `moves\_remaining` by 1.
- An invalid swap that creates no match should be reverted AND should NOT decrease `moves\_remaining`.
- The move counter should therefore only decrease after a successful/valid swap.
- Do not change the maximum number of moves or any other game rules.

Please inspect the existing swap-related code, especially `process\_swap()`, and fix only this bug.

Requirements:

- Make the smallest possible change.
- Preserve the existing matching, swapping, scoring, and board logic.
- Do NOT rewrite the entire `board.py`.
- Do NOT change the UI or add new features.
- Do NOT modify unrelated functions or files.
- Keep my previous fix for the `c` variable in `drop\_and\_refill()` unchanged.
- After making the change, briefly explain why invalid swaps were still consuming a move and exactly what you changed.

code

```
import random
import pygame
```

```
GRID_SIZE = 8
TILE_SIZE = 60
GEM_COLORS = [
    (220, 50, 50), # Red
    (50, 200, 50), # Green
    (50, 100, 240), # Blue
    (240, 200, 40), # Yellow
    (180, 50, 220), # Purple
    (240, 130, 40), # Orange
]
```

class Gem:

```
def __init__(self, color, target_row, col):
    self.color = color
    self.target_row = target_row
    self.col = col
    # Start higher up to animate falling down
    self.current_y = (target_row - 2) * TILE_SIZE
    self.target_y = target_row * TILE_SIZE
```

```

        self.fall_speed = 12.0

    def update(self):
        if self.current_y < self.target_y:
            self.current_y += self.fall_speed
        if self.current_y > self.target_y:
            self.current_y = self.target_y

    def is_animating(self):
        return self.current_y < self.target_y

class Board:
    """Manages animated gem grid, gravity drops, score, and game limits."""

    def __init__(self, offset_x, offset_y, target_score=500, max_moves=20):
        self.offset_x = offset_x
        self.offset_y = offset_y
        self.target_score = target_score
        self.max_moves = max_moves
        self.grid = [[None for _ in range(GRID_SIZE)] for _ in range(GRID_SIZE)]
        self.selected = None
        self.score = 0
        self.moves_remaining = max_moves
        self.reset()

    def reset(self):
        """Reset board grid, score, and move limits."""
        self.score = 0
        self.moves_remaining = self.max_moves
        self.selected = None
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE):
                color = random.choice(GEM_COLORS)
                gem = Gem(color, r, c)
                gem.current_y = gem.target_y # Snap instantly on initial start
                self.grid[r][c] = gem

        self.resolve_matches()

    def is_animating(self):
        """Returns True if any gem is currently dropping down."""
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE):
                if self.grid[r][c] and self.grid[r][c].is_animating():
                    return True
        return False

    def swap_gems(self, pos1, pos2):
        """Swap positions and target render coordinates of two gems."""
        r1, c1 = pos1
        r2, c2 = pos2

        g1, g2 = self.grid[r1][c1], self.grid[r2][c2]
        self.grid[r1][c1], self.grid[r2][c2] = g2, g1

        if self.grid[r1][c1]:
            self.grid[r1][c1].target_row = r1
            self.grid[r1][c1].target_y = r1 * TILE_SIZE

```

```

        self.grid[r1][c1].current_y = r1 * TILE_SIZE

    if self.grid[r2][c2]:
        self.grid[r2][c2].target_row = r2
        self.grid[r2][c2].target_y = r2 * TILE_SIZE
        self.grid[r2][c2].current_y = r2 * TILE_SIZE

    def is_adjacent(self, pos1, pos2):
        r1, c1 = pos1
        r2, c2 = pos2
        return abs(r1 - r2) + abs(c1 - c2) == 1

    def find_matches(self):
        """Scan grid for horizontal and vertical 3-in-a-row color matches."""
        matched = set()

        # Horizontal matches
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE - 2):
                if (
                    self.grid[r][c]
                    and self.grid[r][c + 1]
                    and self.grid[r][c + 2]
                    and self.grid[r][c].color == self.grid[r][c + 1].color == self.grid[r][c
+ 2].color
                ):
                    matched.update([(r, c), (r, c + 1), (r, c + 2)])

        # Vertical matches
        for r in range(GRID_SIZE - 2):
            for c in range(GRID_SIZE):
                if (
                    self.grid[r][c]
                    and self.grid[r + 1][c]
                    and self.grid[r + 2][c]
                    and self.grid[r][c].color == self.grid[r + 1][c].color == self.grid[r +
2][c].color
                ):
                    matched.update([(r, c), (r + 1, c), (r + 2, c)])

        return matched

    def drop_and_refill(self):
        for c in range(GRID_SIZE):
            empty_slots = 0
            for r in range(GRID_SIZE - 1, -1, -1):
                if self.grid[r][c] is None:
                    empty_slots += 1
                elif empty_slots > 0:
                    gem = self.grid[r][c]
                    gem.target_row = r + empty_slots
                    gem.target_y = (r + empty_slots) * TILE_SIZE
                    self.grid[r + empty_slots][c] = gem
                    self.grid[r][c] = None

            for r in range(empty_slots):
                color = random.choice(GEM_COLORS)
                gem = Gem(color, r, c)
                gem.current_y = -(empty_slots - r) * TILE_SIZE

```

```

        self.grid[r][c] = gem

    def resolve_matches(self):
        total_cleared = 0
        while True:
            matches = self.find_matches()
            if not matches:
                break
            total_cleared += len(matches)
            for r, c in matches:
                self.grid[r][c] = None
            self.drop_and_refill()
        return total_cleared

    def process_swap(self, pos1, pos2):
        if not self.is_adjacent(pos1, pos2) or self.is_game_over() or self.is_animating():
            return False

        self.swap_gems(pos1, pos2)
        matches = self.find_matches()

        # BUG SYMPTOM:
        # Move count decrements on EVERY swap attempt even invalid ones.
        self.moves_remaining -= 1

        if not matches:
            self.swap_gems(pos1, pos2) # Revert invalid swap
            return False

        cleared = self.resolve_matches()
        self.score += cleared * 10
        return True

    def is_game_over(self):
        return self.score >= self.target_score or self.moves_remaining <= 0

    def check_result(self):
        if self.score >= self.target_score:
            return "WIN"
        if self.moves_remaining <= 0:
            return "LOSS"
        return None

    def update(self):
        for r in range(GRID_SIZE):
            for c in range(GRID_SIZE):
                if self.grid[r][c]:
                    self.grid[r][c].update()

    def render(self, surface):
        board_rect = pygame.Rect(
            self.offset_x, self.offset_y, GRID_SIZE * TILE_SIZE, GRID_SIZE * TILE_SIZE
        )
        pygame.draw.rect(surface, (20, 22, 28), board_rect, border_radius=8)
        pygame.draw.rect(surface, (60, 65, 75), board_rect, width=3, border_radius=8)

        for r in range(GRID_SIZE):

```

```

for c in range(GRID_SIZE):
    gem = self.grid[r][c]
    if gem:
        x = self.offset_x + c * TILE_SIZE
        y = self.offset_y + gem.current_y
        tile_rect = pygame.Rect(x + 2, y + 2, TILE_SIZE - 4, TILE_SIZE - 4)

        pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)
        pygame.draw.rect(
            surface, (255, 255, 255), tile_rect, width=1, border_radius=10
        )

    if self.selected == (r, c):
        sel_x = self.offset_x + c * TILE_SIZE
        sel_y = self.offset_y + r * TILE_SIZE
        sel_rect = pygame.Rect(sel_x + 2, sel_y + 2, TILE_SIZE - 4,
TILE_SIZE - 4)
        pygame.draw.rect(
            surface, (255, 255, 255), sel_rect, width=4, border_radius=10
        )

```

9/30/2026, 7:47:20 PM



The bug is in `process_swap()`. The move is being decremented **before** the code checks whether the swap created a match.

Currently:

```

self.swap_gems(pos1, pos2)
matches = self.find_matches()

self.moves_remaining -= 1 # happens for every swap

if not matches:
    self.swap_gems(pos1, pos2)
    return False

```

So even an invalid swap reaches `self.moves_remaining -= 1`.

Minimal fix

Move this one line:

```
self.moves_remaining -= 1
```

to **after** the `if not matches` check.

Your corrected `process_swap()` should be:

```

def process_swap(self, pos1, pos2):
    if not self.is_adjacent(pos1, pos2) or self.is_game_over() or self.is_animating():
        return False

    self.swap_gems(pos1, pos2)
    matches = self.find_matches()

    if not matches:
        self.swap_gems(pos1, pos2) # Revert invalid swap

```

```
        return False

    self.moves_remaining -= 1

    cleared = self.resolve_matches()
    self.score += cleared * 10
    return True
```

Exactly what changed

Before:

```
matches = self.find_matches()
self.moves_remaining -= 1

if not matches:
```

After:

```
matches = self.find_matches()

if not matches:
    self.swap_gems(pos1, pos2)
    return False

self.moves_remaining -= 1
```

That's the **only change needed**.

The resulting behavior is:

Swap	Match?	Revert?	Move decreases?
Valid swap	Yes	No	✔ Yes
Invalid swap	No	✔ Yes	✘ No

Your previous `drop_and_refill()` fix remains completely unchanged. No other functions or files need to be modified.

9/30/2026, 7:47:22 PM · gpt-5-6



okay now for the same code
I am working on a Python Pygame Match-3 game.

I have already fixed two bugs:

1. The `c` variable NameError in `drop\_and\_refill()`.
2. Invalid swaps incorrectly consuming a move.

Now implement the next required feature:

****Cascade Scoring****

When a valid swap creates a match, the game should score matches based on their cascade level:

- First match in a move: 1× score
- Second consecutive match caused by the refill/drop: 2× score
- Third consecutive match: 3× score
- Fourth consecutive match: 4× score
- Continue increasing the multiplier for every subsequent cascade.

Example:

If the normal score for a match is 10:

- First match = 10 points
- Second cascade = 20 points
- Third cascade = 30 points

The cascade multiplier must reset to 1× when the player makes a new swap.

Please inspect the existing matching, scoring, and `resolve\_matches()` logic before making changes.

Requirements:

- Make the smallest possible changes.
- Preserve the existing game behavior.
- Do NOT rewrite the entire `board.py`.
- Do NOT change the existing scoring formula except to apply the cascade multiplier.
- Do NOT change move counting.
- Do NOT change swap validation.
- Do NOT add unrelated features.
- Keep my previous fixes unchanged.
- If possible, implement the multiplier within the existing match-resolution flow rather than creating an unnecessarily complicated new system.
- After making the change, briefly explain which functions were changed and how the cascade multiplier works.

9/30/2026, 7:51:52 PM



Thinking

•
|

✓ Worked for 9s

Yes. The existing structure already makes this fairly simple because `resolve_matches()` is the function that handles the entire chain of matches and refills.

The cleanest small change is to let `resolve_matches()` apply cascade scoring when called after a player swap, while keeping the initial `reset()` resolution from awarding points.

1. Change `resolve_matches()`

Replace only this function:

```
def resolve_matches(self, score_cascades=False):
    total_cleared = 0
    cascade_multiplier = 1

    while True:
        matches = self.find_matches()
        if not matches:
            break

        total_cleared += len(matches)

        if score_cascades:
            self.score += len(matches) * 10 * cascade_multiplier

        for r, c in matches:
            self.grid[r][c] = None

        self.drop_and_refill()
        cascade_multiplier += 1

    return total_cleared
```

2. Change `process_swap()`

Your current corrected `process_swap()` has:

```
cleared = self.resolve_matches()
self.score += cleared * 10
return True
```

Change only those lines to:

```
self.resolve_matches(score_cascades=True)
return True
```

So the complete function becomes:

```
def process_swap(self, pos1, pos2):
    if not self.is_adjacent(pos1, pos2) or self.is_game_over() or self.is_animating():
        return False

    self.swap_gems(pos1, pos2)
    matches = self.find_matches()

    if not matches:
        self.swap_gems(pos1, pos2) # Revert invalid swap
```

```

        return False

    self.moves_remaining -= 1

    self.resolve_matches(score_cascades=True)
    return True

```

How the cascade multiplier works

Inside `resolve_matches()`:

```
cascade_multiplier = 1
```

starts every resolution at **1x**.

Suppose the first match clears 3 gems:

```
3 gems × 10 × 1 = 30
```

The gems fall and create another match:

```
3 gems × 10 × 2 = 60
```

Another refill creates a third match:

```
3 gems × 10 × 3 = 90
```

So:

```

First match    → 1×
Second match   → 2×
Third match    → 3×
Fourth match   → 4×
...

```

The important part is that:

```
cascade_multiplier = 1
```

is created **inside** `resolve_matches()`. Therefore, every new call to `resolve_matches()` starts at 1x. A new player swap automatically resets the cascade multiplier.

Why `score_cascades=False` is needed

Your `reset()` currently does:

```
self.resolve_matches()
```

The board can randomly contain matches when it is created. Those matches are removed during initialization, but they **shouldn't give the player points**.

Because the default is:

```
score_cascades=False
```

the reset still removes those matches without increasing the score.

Your actual player swap does:

```
self.resolve_matches(score_cascades=True)
```

so only player-triggered match resolution awards points.

Only `resolve_matches()` **and the scoring part of** `process_swap()` **are changed.** Your `c` fix in `drop_and_refill()`, invalid-swap fix, move counting, matching, swapping, and UI remain unchanged.

9/30/2026, 7:52:03 PM · gpt-5-6



I am working on a Python Pygame Match-3 game.

I have already completed these tasks:

1. Fixed the `c` variable NameError in `drop\_and\_refill()`.
2. Fixed invalid swaps incorrectly consuming a move.
3. Added cascade scoring with 1x, 2x, 3x, etc. multipliers.

Now implement the next required feature:

****Bomb Gem for 4-in-a-row matches****

When the player creates a match of exactly 4 gems in a row:

- A special Bomb Gem should be created at an appropriate position in the matched group.
- The Bomb Gem must be visually distinguishable from normal gems, preferably with a glow, outline, or other clear visual effect.
- When the Bomb Gem is activated as part of a later match/swap, it should clear the entire row OR entire column in which it is activated.
- The normal match and existing scoring/cascade behavior should continue to work.
- Existing 3-gem matches should continue behaving normally.
- Existing cascade scoring must remain unchanged.

Please inspect the existing matching, clearing, refill, drawing/rendering, and swap logic before making changes.

Requirements:

- Make the smallest reasonable changes.
- Preserve all existing functionality.
- Do NOT rewrite the entire `board.py`.
- Do NOT remove or replace my previous fixes.
- Do NOT change the existing move-counting behavior.
- Do NOT change the cascade multiplier behavior.
- Do NOT add unrelated features.
- Keep the implementation compatible with the existing Pygame code.
- After making the changes, briefly explain:
 1. How a 4-gem match is detected.
 2. How the Bomb Gem is represented.
 3. How it is drawn differently.
 4. How activating it clears a row or column.

9/30/2026, 9:05:54 PM



Yes. Since you already have the `c` fix, invalid-swap fix, and cascade scoring, we can add the Bomb Gem with a few targeted changes.

The main complication is that your current `find_matches()` returns only a `set` of matched positions. To detect **exactly 4 in a row** and know whether it was horizontal or vertical, we need to preserve that information during match resolution.

I would make these changes only.

1. Update `Gem` to support Bomb Gems

In `Gem.__init__()`, add one property:

```
self.is_bomb = False
self.bomb_direction = None
```

So the beginning becomes:

```
class Gem:

    def __init__(self, color, target_row, col):
        self.color = color
        self.target_row = target_row
        self.col = col

        self.is_bomb = False
        self.bomb_direction = None

        # Start higher up to animate falling down
        self.current_y = (target_row - 2) * TILE_SIZE
        self.target_y = target_row * TILE_SIZE
        self.fall_speed = 12.0
```

For a normal gem:

```
is_bomb = False
```

For a Bomb Gem:

```
is_bomb = True
bomb_direction = "row"
```

or:

```
bomb_direction = "column"
```

2. Replace `find_matches()`

Your current function only returns the matched cells. Replace it with this version:

```
def find_matches(self):
    """Scan grid for horizontal and vertical matches."""
    matched = set()
    bomb_matches = []

    # Horizontal matches
    for r in range(GRID_SIZE):
        c = 0
        while c < GRID_SIZE:
            if self.grid[r][c] is None:
                c += 1
                continue

            start = c
            color = self.grid[r][c].color

            while (
                c < GRID_SIZE
```

```

        and self.grid[r][c] is not None
        and self.grid[r][c].color == color
    ):
        c += 1

    length = c - start

    if length >= 3:
        for col in range(start, c):
            matched.add((r, col))

            if length == 4:
                bomb_matches.append(
                    ((r, start + 1), "row")
                )

# Vertical matches
for c in range(GRID_SIZE):
    r = 0
    while r < GRID_SIZE:
        if self.grid[r][c] is None:
            r += 1
            continue

        start = r
        color = self.grid[r][c].color

        while (
            r < GRID_SIZE
            and self.grid[r][c] is not None
            and self.grid[r][c].color == color
        ):
            r += 1

        length = r - start

        if length >= 3:
            for row in range(start, r):
                matched.add((row, c))

            if length == 4:
                bomb_matches.append(
                    ((start + 1, c), "column")
                )

    return matched, bomb_matches

```

What this changes

It now detects runs such as:



and records:

```
((row, middle_position), "row")
```

For a vertical match:



it records:

```
((middle_position, column), "column")
```

The Bomb is placed at the **second gem of the four**. That's a simple deterministic position inside the matched group. Importantly, a run of 5 does **not** create a Bomb because the requirement is **exactly 4**.

3. Update `resolve_matches()`

Because `find_matches()` now returns two things, change `resolve_matches()` to:

```
def resolve_matches(self, score_cascades=False):
    total_cleared = 0
    cascade_multiplier = 1

    while True:
        matches, bomb_matches = self.find_matches()

        if not matches:
            break

        total_cleared += len(matches)

        if score_cascades:
            self.score += len(matches) * 10 * cascade_multiplier

        # Create Bomb Gems before clearing the matched gems.
        for (r, c), direction in bomb_matches:
            if self.grid[r][c] is not None:
                self.grid[r][c].is_bomb = True
                self.grid[r][c].bomb_direction = direction

        # Check for Bomb Gems being activated.
        activated_bombs = []

        for r, c in matches:
            gem = self.grid[r][c]

            if gem is not None and gem.is_bomb:
                activated_bombs.append((r, c, gem.bomb_direction))

        # Clear the normal matched gems.
        for r, c in matches:
            self.grid[r][c] = None

        # Bomb activation clears an entire row or column.
        for r, c, direction in activated_bombs:
            if direction == "row":
                for col in range(GRID_SIZE):
                    self.grid[r][col] = None

            elif direction == "column":
```

```

        for row in range(GRID_SIZE):
            self.grid[row][c] = None

        self.drop_and_refill()
        cascade_multiplier += 1

    return total_cleared

```

One important detail

Your existing cascade scoring stays here:

```
self.score += len(matches) * 10 * cascade_multiplier
```

So the multiplier behavior is unchanged:

```

1st cascade → ×1
2nd cascade → ×2
3rd cascade → ×3
...

```

And your existing:

```
resolve_matches()
```

during `reset()` still works because `score_cascades` defaults to `False`.

4. Update the rendering

Inside `render()`, you currently draw every gem like this:

```

pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)
pygame.draw.rect(
    surface, (255, 255, 255), tile_rect, width=1, border_radius=10
)

```

Replace only that section with:

```

pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)

if gem.is_bomb:
    # Bomb glow/outline
    pygame.draw.rect(
        surface,
        (255, 255, 255),
        tile_rect.inflate(-8, -8),
        width=4,
        border_radius=10,
    )

    # Direction indicator
    center_x = tile_rect.centerx
    center_y = tile_rect.centery

    if gem.bomb_direction == "row":
        pygame.draw.line(
            surface,

```

```

        (255, 255, 255),
        (tile_rect.left + 8, center_y),
        (tile_rect.right - 8, center_y),
        width=4,
    )
elif gem.bomb_direction == "column":
    pygame.draw.line(
        surface,
        (255, 255, 255),
        (center_x, tile_rect.top + 8),
        (center_x, tile_rect.bottom - 8),
        width=4,
    )
else:
    pygame.draw.rect(
        surface,
        (255, 255, 255),
        tile_rect,
        width=1,
        border_radius=10,
    )

```

So a Bomb Gem gets a thick white outline and a direction indicator.

5. Important: update `find_matches()` callers

Because `find_matches()` now returns:

```
matches, bomb_matches
```

there is one existing line in `process_swap()` that needs to change.

Currently:

```
matches = self.find_matches()
```

Change it to:

```
matches, _ = self.find_matches()
```

So your existing swap logic remains essentially identical:

```

def process_swap(self, pos1, pos2):
    if not self.is_adjacent(pos1, pos2) or self.is_game_over() or self.is_animating():
        return False

    self.swap_gems(pos1, pos2)
    matches, _ = self.find_matches()

    if not matches:
        self.swap_gems(pos1, pos2)
        return False

    self.moves_remaining -= 1

    self.resolve_matches(score_cascades=True)
    return True

```

Your **invalid swap fix stays unchanged**.

How the Bomb works

1. Detecting four

A sequence:



has length exactly 4, so:

```
if length == 4:
```

creates a Bomb.

A sequence of 3:



doesn't create one.

A sequence of 5:



also doesn't create one under your requested **exactly-4** rule.

2. Bomb representation

Every Gem gets:

```
self.is_bomb = False
self.bomb_direction = None
```

A generated Bomb becomes:

```
is_bomb = True
bomb_direction = "row"
```

or:

```
is_bomb = True
bomb_direction = "column"
```

3. Visual appearance

Normal gems retain their existing appearance.

Bombs get:

- thicker white outline
- horizontal indicator for a row Bomb
- vertical indicator for a column Bomb

4. Activation

If a Bomb becomes part of a later match, `resolve_matches()` detects it:

```
if gem is not None and gem.is_bomb:
```

Then:

```
direction == "row"
```

clears its entire row.

Or:

```
direction == "column"
```

clears its entire column.

One limitation to be aware of

With the existing game architecture, this implementation activates a Bomb **when it is included in a later match**. It does not make a Bomb independently explode merely because the player swaps it with an adjacent gem that doesn't otherwise form a match.

That keeps the implementation small and fits your requirement that the Bomb be activated **as part of a later match/swap**, without changing your existing swap rules or adding a separate activation system.

9/30/2026, 9:06:00 PM · gpt-5-6



```

C:\Users\sufiy\57_match_three_gem>python main.py
pygame 2.6.1 (SDL 2.28.4, Python 3.13.1)
Hello from the pygame community. https://www.pygame.org/contribute.html
Traceback (most recent call last):
  File "C:\Users\sufiy\57_match_three_gem\main.py", line 38, in <module>
    main()
    ~~~~^^
  File "C:\Users\sufiy\57_match_three_gem\main.py", line 16, in main
    engine = GameEngine(SCREEN_WIDTH, SCREEN_HEIGHT)
  File "C:\Users\sufiy\57_match_three_gem\game\game_engine.py", line 12, in
__init__
    self.board = Board(offset_x, offset_y, target_score=500, max_moves=20)
    ~~~~~
  File "C:\Users\sufiy\57_match_three_gem\game\board.py", line 53, in __init__
    self.reset()
    ~~~~~
  File "C:\Users\sufiy\57_match_three_gem\game\board.py", line 67, in reset
    self.resolve_matches()
    ~~~~~
  File "C:\Users\sufiy\57_match_three_gem\game\board.py", line 198, in
resolve_matches
    for r, c in matches:
      ^^^^
ValueError: not enough values to unpack (expected 2, got 0)

```

9/30/2026, 9:13:12 PM



Yep — this error is caused by the change I suggested to `find_matches()`.

The problem is that we changed:

```
return matched, bomb_matches
```

but `resolve_matches()` still has code that expects `matches` to contain `(r, c)` pairs. During `reset()`, it is now getting the wrong structure.

The smallest and safest fix is to **not change the return structure of `find_matches()` at all**. Instead, keep `find_matches()` returning the original `set` of positions and separately detect 4-in-a-row inside `resolve_matches()`.

That avoids breaking your existing matching logic.

Fix

Restore your original `find_matches()` exactly as it was.

Then use this `resolve_matches()`:

```

def resolve_matches(self, score_cascades=False):
    total_cleared = 0
    cascade_multiplier = 1

    while True:
        matches = self.find_matches()

        if not matches:
            break

```

```

total_cleared += len(matches)

if score_cascades:
    self.score += len(matches) * 10 * cascade_multiplier

# Find exactly-4 matches and create bombs.
bomb_matches = []

# Horizontal
for r in range(GRID_SIZE):
    for c in range(GRID_SIZE - 3):
        if (
            self.grid[r][c]
            and self.grid[r][c + 1]
            and self.grid[r][c + 2]
            and self.grid[r][c + 3]
            and self.grid[r][c].color
            == self.grid[r][c + 1].color
            == self.grid[r][c + 2].color
            == self.grid[r][c + 3].color
        ):
            bomb_matches.append((r, c + 1), "row")

# Vertical
for r in range(GRID_SIZE - 3):
    for c in range(GRID_SIZE):
        if (
            self.grid[r][c]
            and self.grid[r + 1][c]
            and self.grid[r + 2][c]
            and self.grid[r + 3][c]
            and self.grid[r][c].color
            == self.grid[r + 1][c].color
            == self.grid[r + 2][c].color
            == self.grid[r + 3][c].color
        ):
            bomb_matches.append((r + 1, c), "column")

# Create Bomb Gems before clearing the match.
for (r, c), direction in bomb_matches:
    if self.grid[r][c] is not None:
        self.grid[r][c].is_bomb = True
        self.grid[r][c].bomb_direction = direction

# Find bombs that are part of this match.
activated_bombs = []

for r, c in matches:
    gem = self.grid[r][c]

    if gem is not None and gem.is_bomb:
        activated_bombs.append((r, c, gem.bomb_direction))

# Clear normal matched gems.
for r, c in matches:
    self.grid[r][c] = None

# Clear rows/columns from activated bombs.

```

```

    for r, c, direction in activated_bombs:
        if direction == "row":
            for col in range(GRID_SIZE):
                self.grid[r][col] = None

        elif direction == "column":
            for row in range(GRID_SIZE):
                self.grid[row][c] = None

    self.drop_and_refill()
    cascade_multiplier += 1

    return total_cleared

```

And in `process_swap()`, keep your previous fix:

```

def process_swap(self, pos1, pos2):
    if not self.is_adjacent(pos1, pos2) or self.is_game_over() or self.is_animating():
        return False

    self.swap_gems(pos1, pos2)
    matches = self.find_matches()

    if not matches:
        self.swap_gems(pos1, pos2)
        return False

    self.moves_remaining -= 1

    self.resolve_matches(score_cascades=True)
    return True

```

Why this fixes your current crash

Your original architecture expects:

```
matches = self.find_matches()
```

to produce:

```
{(row, column), (row, column), ...}
```

Then this works:

```
for r, c in matches:
```

The previous implementation changed it into something like:

```
(matches, bomb_matches)
```

which changed the interface of `find_matches()` and broke existing code.

The corrected approach **keeps** `find_matches()` **unchanged** and detects the 4-gem pattern inside `resolve_matches()` instead.

So your existing:

- `drop_and_refill()` fix 

- invalid-swap fix 
- cascade scoring 
- original matching logic 
- move counting 

remain intact.

One thing: make sure your `Gem.__init__()` still contains:

```
self.is_bomb = False
self.bomb_direction = None
```

That part of the previous change should remain.

9/30/2026, 9:13:14 PM · gpt-5-6



Pasted code.py

I am working on a Python Pygame Match-3 game.

I have already completed these tasks:

1. Fixed the `c` variable NameError in `drop\_and\_refill()`.
2. Fixed invalid swaps incorrectly consuming a move.
3. Added cascade scoring with 1x, 2x, 3x, etc. multipliers.
4. Added a Bomb Gem for 4-in-a-row matches.

Now implement the final required feature:

****Idle Hint After 5 Seconds****

If the player has not made a move for more than 5 seconds:

- Find a valid adjacent gem swap that would create a match.
- Visually highlight the two gems involved in that valid swap.
- The highlight should be noticeable, such as a pulsing, shimmering, or glowing effect.
- The game must NOT automatically perform the swap.
- As soon as the player makes a move, the idle timer should reset and the hint should disappear.
- If there is no valid swap available, do not show a hint.

Please inspect the existing board, swap-validation, update/game-loop, and drawing/rendering logic before making changes.

Requirements:

- Make the smallest reasonable changes.
- Preserve all existing functionality.
- Do NOT rewrite the entire `board.py` or other files.
- Do NOT change the existing matching or scoring logic.
- Do NOT change move counting.
- Do NOT change Bomb Gem behavior.
- Do NOT change cascade scoring.
- Keep all previous fixes unchanged.
- Do NOT automatically swap or move gems for the player.
- Use the existing Pygame timing/game-loop system if one already exists.
- After making the changes, briefly explain:
 1. How the 5-second idle timer works.
 2. How a valid hint swap is found.
 3. How the two gems are highlighted.
 4. How the hint is cleared/reset after the player moves.

here is the current code

9/30/2026, 9:22:22 PM



I inspected the current `board.py`. The existing game loop already calls `Board.update()` and `Board.render()`, so we can implement the idle hint entirely inside `Board` without changing `main.py` or `game_engine.py`. Your existing swap, scoring, cascade, and Bomb logic can remain intact. `Pasted code.py`



Make these **small changes only**.

1. Add idle/hint state in `__init__()`

Right after:

```
self.moves_remaining = max_moves
```

add:

```
self.idle_start_time = pygame.time.get_ticks()
self.hint_positions = None
self.hint_check_time = 0
```

So:

```
self.score = 0
self.moves_remaining = max_moves

self.idle_start_time = pygame.time.get_ticks()
self.hint_positions = None
self.hint_check_time = 0

self.reset()
```

2. Reset the timer when the board resets

In `reset()`, after:

```
self.selected = None
```

add:

```
self.idle_start_time = pygame.time.get_ticks()
self.hint_positions = None
self.hint_check_time = 0
```

Your existing reset logic stays unchanged. `Pasted code.py`

3. Add a function to find a valid hint swap

Add this function before `process_swap()` :

```
def find_hint_swap(self):
    """Find an adjacent swap that creates a match."""
    for r in range(GRID_SIZE):
        for c in range(GRID_SIZE):
            # Try swapping right
            if c < GRID_SIZE - 1:
                pos1 = (r, c)
                pos2 = (r, c + 1)

                self.swap_gems(pos1, pos2)
                matches = self.find_matches()
                self.swap_gems(pos1, pos2)

            if matches:
```

```

        return pos1, pos2

    # Try swapping down
    if r < GRID_SIZE - 1:
        pos1 = (r, c)
        pos2 = (r + 1, c)

        self.swap_gems(pos1, pos2)
        matches = self.find_matches()
        self.swap_gems(pos1, pos2)

        if matches:
            return pos1, pos2

    return None

```

This tests possible swaps without actually performing one.

For example:

```

A B C D
E F G H

```

It temporarily tries:

```

A ↔ B
A ↔ E
B ↔ C
B ↔ F
...

```

and checks whether the resulting board has a match.

If it finds one, it returns the two positions.

4. Update the existing `update()` function

Your current `update()` only updates falling gems. [Pasted code.py](#)

Replace it with:

```

def update(self):
    for r in range(GRID_SIZE):
        for c in range(GRID_SIZE):
            if self.grid[r][c]:
                self.grid[r][c].update()

    # Do not show hints while gems are falling.
    if self.is_animating():
        self.hint_positions = None
        self.idle_start_time = pygame.time.get_ticks()
        return

    current_time = pygame.time.get_ticks()

    # Player has been idle for more than 5 seconds.
    if current_time - self.idle_start_time >= 5000:
        if self.hint_positions is None:
            # Avoid repeatedly searching the board every frame.
            if current_time - self.hint_check_time >= 1000:

```

```
self.hint_check_time = current_time
self.hint_positions = self.find_hint_swap()
```

The `5000` means **5000 milliseconds = 5 seconds**.

5. Reset the hint when a valid move is made

Your current `process_swap()` is already correct regarding move counting:

```
if not matches:
    self.swap_gems(pos1, pos2)
    return False

self.moves_remaining -= 1
```

Keep that exactly as it is. `Pasted code.py`

Immediately after:

```
self.moves_remaining -= 1
```

add:

```
self.idle_start_time = pygame.time.get_ticks()
self.hint_positions = None
self.hint_check_time = 0
```

So that section becomes:

```
if not matches:
    self.swap_gems(pos1, pos2)
    return False

self.moves_remaining -= 1

self.idle_start_time = pygame.time.get_ticks()
self.hint_positions = None
self.hint_check_time = 0

self.resolve_matches(score_cascades=True)
return True
```

This means the hint disappears immediately after a **successful player move**, and the 5-second timer starts again.

6. Add the visual hint to `render()`

Your existing gem drawing is here:

```
pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)
pygame.draw.rect(
    surface, (255, 255, 255), tile_rect, width=1, border_radius=10
)
```

Keep that.

Then, **after the normal gem drawing and before the existing** `self.selected` **block**, add:

```
if self.hint_positions and (r, c) in self.hint_positions:
    pulse = (pygame.time.get_ticks() // 150) % 2
```

```

if pulse:
    pygame.draw.rect(
        surface,
        (255, 255, 255),
        tile_rect.inflate(8, 8),
        width=4,
        border_radius=12,
    )

```

So the relevant part becomes:

```

pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)
pygame.draw.rect(
    surface, (255, 255, 255), tile_rect, width=1, border_radius=10
)

if self.hint_positions and (r, c) in self.hint_positions:
    pulse = (pygame.time.get_ticks() // 150) % 2

    if pulse:
        pygame.draw.rect(
            surface,
            (255, 255, 255),
            tile_rect.inflate(8, 8),
            width=4,
            border_radius=12,
        )

if self.selected == (r, c):
    ...

```

This gives the two hint gems a **pulsing outline**.

What the final behavior is Immediately after a move

```

Player makes valid swap
    ↓
Hint disappears
    ↓
Timer resets to 0
    ↓
Normal matching/scoring/cascade happens

```

After 5 seconds

```

No player move for 5 seconds
    ↓
Find possible adjacent swaps
    ↓
Find first swap that creates a match
    ↓
Highlight those 2 gems

```

The game **does not perform the swap**.

If there is no possible move

```
5 seconds idle
↓
No valid swap found
↓
No hint shown
```

Your previous features remain untouched

- `drop_and_refill()` c fix  Pasted code.py
- Invalid swaps don't consume moves 
- Cascade scoring remains `10 × cascade_multiplier`  Pasted code.py
- Bomb Gems remain unchanged  Pasted code.py
- No automatic swapping 
- No changes to maximum moves 
- No changes to the existing matching rules 

One small implementation detail: the hint search uses your existing `swap_gems()` temporarily, then immediately swaps the gems back, so it reuses your existing swap/match logic rather than creating a second matching algorithm.

9/30/2026, 9:22:25 PM · gpt-5-6



is the indexing correct?

```
def render(self, surface):
    board_rect = pygame.Rect(
        self.offset_x, self.offset_y, GRID_SIZE * TILE_SIZE, GRID_SIZE * TILE_SIZE
    )
    pygame.draw.rect(surface, (20, 22, 28), board_rect, border_radius=8)
    pygame.draw.rect(surface, (60, 65, 75), board_rect, width=3,
border_radius=8)

    for r in range(GRID_SIZE):
        for c in range(GRID_SIZE):
            gem = self.grid[r][c]
            if gem:
                x = self.offset_x + c * TILE_SIZE
                y = self.offset_y + gem.current_y
                tile_rect = pygame.Rect(x + 2, y + 2, TILE_SIZE - 4, TILE_SIZE - 4)

                pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)
                pygame.draw.rect(
                    surface, (255, 255, 255), tile_rect, width=1, border_radius=10
                )

            if self.hint_positions and (r, c) in self.hint_positions:
                pulse = (pygame.time.get_ticks() // 150) % 2

                if pulse:
                    pygame.draw.rect(
                        surface,
                        (255, 255, 255),
                        tile_rect.inflate(8, 8),
                        width=4,
                        border_radius=12,
                    )

            if self.selected == (r, c):
                sel_x = self.offset_x + c * TILE_SIZE
                sel_y = self.offset_y + r * TILE_SIZE
                sel_rect = pygame.Rect(sel_x + 2, sel_y + 2, TILE_SIZE - 4,
TILE_SIZE - 4)
                pygame.draw.rect(
                    surface, (255, 255, 255), sel_rect, width=4, border_radius=10
                )
```

9/30/2026, 9:32:53 PM



Almost — **the** `(r, c)` **indexing is correct**, but there is one important issue with `tile_rect`.

Your indexing

This is correct:

```
for r in range(GRID_SIZE):
    for c in range(GRID_SIZE):
        gem = self.grid[r][c]
```



So:

- `r` = row → vertical position
- `c` = column → horizontal position

And this is also correct:

```
if self.hint_positions and (r, c) in self.hint_positions:
```

because your hint positions are stored as `(row, column)` pairs.

The problem is `tile_rect`

You create `tile_rect` **only inside**:

```
if gem:
    ...
    tile_rect = pygame.Rect(...)
```

but then use it outside that `if`:

```
if self.hint_positions and (r, c) in self.hint_positions:
    ...
    tile_rect.inflate(8, 8)
```

If `gem` is `None`, `tile_rect` won't be created for that iteration. That can cause an error or, worse, use a stale value from the previous iteration.

Fix

Put the hint code **inside** `if gem:`, like this:

```
for r in range(GRID_SIZE):
    for c in range(GRID_SIZE):
        gem = self.grid[r][c]
        if gem:
            x = self.offset_x + c * TILE_SIZE
            y = self.offset_y + gem.current_y
            tile_rect = pygame.Rect(
                x + 2, y + 2, TILE_SIZE - 4, TILE_SIZE - 4
            )

            pygame.draw.rect(
                surface, gem.color, tile_rect, border_radius=10
            )
            pygame.draw.rect(
                surface, (255, 255, 255), tile_rect,
                width=1, border_radius=10
            )

            if self.hint_positions and (r, c) in self.hint_positions:
                pulse = (pygame.time.get_ticks() // 150) % 2

                if pulse:
                    pygame.draw.rect(
                        surface,
                        (255, 255, 255),
                        tile_rect.inflate(8, 8),
```

```

        width=4,
        border_radius=12,
    )

    if self.selected == (r, c):
        sel_x = self.offset_x + c * TILE_SIZE
        sel_y = self.offset_y + r * TILE_SIZE
        sel_rect = pygame.Rect(
            sel_x + 2, sel_y + 2,
            TILE_SIZE - 4, TILE_SIZE - 4
        )
        pygame.draw.rect(
            surface,
            (255, 255, 255),
            sel_rect,
            width=4,
            border_radius=10
        )

```

One more thing: your **hint position indexing is correct**, but your **hint rectangle's y uses** `gem.current_y` **while selection uses** `r * TILE_SIZE`. That's actually reasonable: the hint follows the gem while it is animating, whereas the selection stays on the grid cell.

So the short answer: `(r, c)` **is correct. Just move the hint block inside** `if gem:`.

9/30/2026, 9:32:56 PM · gpt-5-6



now ?

```
def render(self, surface):
    board_rect = pygame.Rect(
        self.offset_x, self.offset_y, GRID_SIZE * TILE_SIZE, GRID_SIZE * TILE_SIZE
    )
    pygame.draw.rect(surface, (20, 22, 28), board_rect, border_radius=8)
    pygame.draw.rect(surface, (60, 65, 75), board_rect, width=3,
border_radius=8)

    for r in range(GRID_SIZE):
        for c in range(GRID_SIZE):
            gem = self.grid[r][c]
            if gem:
                x = self.offset_x + c * TILE_SIZE
                y = self.offset_y + gem.current_y
                tile_rect = pygame.Rect(x + 2, y + 2, TILE_SIZE - 4, TILE_SIZE - 4)

                pygame.draw.rect(surface, gem.color, tile_rect, border_radius=10)
                pygame.draw.rect(
                    surface, (255, 255, 255), tile_rect, width=1, border_radius=10
                )

                if self.hint_positions and (r, c) in self.hint_positions:
                    pulse = (pygame.time.get_ticks() // 150) % 2

                    if pulse:
                        pygame.draw.rect(
                            surface,
                            (255, 255, 255),
                            tile_rect.inflate(8, 8),
                            width=4,
                            border_radius=12,
                        )

                if self.selected == (r, c):
                    sel_x = self.offset_x + c * TILE_SIZE
                    sel_y = self.offset_y + r * TILE_SIZE
                    sel_rect = pygame.Rect(sel_x + 2, sel_y + 2, TILE_SIZE - 4,
TILE_SIZE - 4)
                    pygame.draw.rect(
                        surface, (255, 255, 255), sel_rect, width=4, border_radius=10
                    )
```

9/30/2026, 9:35:16 PM



Yes — **now the indexing and placement are correct.** ✅

Your structure is right:

```
gem = self.grid[r][c]
```

and:

```
if self.hint_positions and (r, c) in self.hint_positions:
```

Both use the same (row, column) = (r, c) convention.

Also, putting the hint code inside:

```
if gem:
```

is correct because `tile_rect` is guaranteed to exist before you use it.

Your flow is now

```
for each row r
  for each column c
    get gem[r][c]

    if gem exists:
      draw gem
      if (r,c) is a hint:
        draw pulsing highlight

    if (r,c) is selected:
      draw selection
```

So **you don't need to change anything in this** `render()` **code.** 👍

One small distinction: the hint follows `gem.current_y`, while selection uses the fixed grid position `r * TILE_SIZE`. That's fine and intentional for animated gems.

9/30/2026, 9:35:19 PM · gpt-5-6